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Probabilistic Combinatorics SS24

Sheet 9

Due: 17. June at 19:00

(1) Let L be a finite distributive lattice, $\mu_1, \mu_2 : L \to \mathbb{R}$ functions satisfying

$$\mu_1(x), \mu_2(x) \geq 0, \text{ for } x \in L$$

$$\mu_1(x)\mu_2(y) \leq \mu_1(x \vee y)\mu_2(x \wedge y) \text{ for } x, y \in L$$

and

$$\sum_{x \in L} \mu_1(x) = \sum_{x \in L} \mu_2(x).$$

Show that for any monotonically increasing function $f : L \to \mathbb{R}$ it holds that

$$\sum_{x \in L} f(x)\mu_1(x) \geq \sum_{x \in L} f(x)\mu_2(x).$$

(2) Let $L_{n,m} = [n]^m$ be the set of all m-tupels of elements of $[n]$. Let

$$(s_1, ..., s_m) \leq (s'_1, ..., s'_m)$$

for $(s_1, ..., s_m), (s'_1, ..., s'_m) \in [n]^m$, iff $s_i \leq s'_i$ for all i.

Show:

(i) $L_{n,m}$ is a distributive lattice. Describe the $\vee$ and $\wedge$ operations in $L_{n,m}$.

(ii) The subset $S_{n,m} \subseteq L_{n,m}$ of all $(s_1, ..., s_m)$ with $s_1 < ... < s_m$ equipped with the same order is also a distributive lattice.

(iii) Let now $n \geq m$. There is a bijection $\phi : S_{n,m} \to S_{n,n-m}$, s.t. for all $x, y \in S_{n,m}$, $\phi(y) \leq \phi(x)$ if and only if $x \leq y$.

(3) Consider the following experiment: For $1 \leq m \leq n$ there are m balls thrown one by one into n baskets. In which baskets the balls land are independent and the j-th ball will land with probability $p_{a_i,j}$ in the basket a_i.

(i) Define the probability space and identify the outcomes with $L_{n,m}$ from (2).

(ii) For $I \subseteq [m]$ and $t_I \in \mathbb{R}$ define the event

$$A_I := \{(a_1, ..., a_m) \in L_{n,m} \mid \sum_{i \in I} a_i \geq t_I\}.$$

Show that, for $I, J \subseteq [m]$ with $I \cap J = \emptyset$ or $I \cup J = [m]$ and real numbers t_I and t_J it holds that

$$\mathbb{P}(A_I \cap A_J) \geq \mathbb{P}(A_I)\mathbb{P}(A_J).$$

(4) A subset $\mathcal{A} \subseteq 2^{[n]}$ is called intersecting, if $A \cap B \neq \emptyset$ for all $A, B \in \mathcal{A}$. Proof: Let $\mathcal{A}_1, ..., \mathcal{A}_k$ be intersecting subsets of $2^{[n]}$, then

$$\#\bigcup_{i=1}^{k} \mathcal{A}_i \leq 2^n - 2^{n-k}.$$